

Cyrptic Exon QPCR

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2024年12月3日 星期二

Old Protocol

RNA extraction with PureLink™ RNA Mini Kit (quick protocol)

Lysis and Homogenization

1. Prepare lysis buffer: 3000uL lysis buffer + 30 uL 2-mercaptoethanol (100x)
2. Take the plate out and plate it on ice
3. Add 300uL lysis buffer for each well and plate the whole ice bucket on the shaker for 15min
4. Collect the lysis buffer into 1.5mL tubes
5. Combine column 1 and column2

240603EB PLATE B plastic plate (1x = 200K)					^
	1	2	3	4	
A	5319 2x	5319 1x	4517 1x		
B	5319 2x	5319 1x	4517 1x		
C	5319 2x	5319 1x	4517 1x		

6. Homogenization: Pass the lysate 5–10 times through an 18- to 21-gauge syringe needle.

RNA Purification

1. Add one volume 70% ethanol to each volume of cell homogenate.
 - 600 uL for 5319
 - 300 uL for 4517
2. Vortex to mix thoroughly and to disperse any visible precipitate that may form after adding ethanol.
3. Transfer up to 700 µL of the sample (including any remaining precipitate) to the spin cartridge (with the collection tube).
4. Centrifuge at 12,000 × g for 30 seconds at room temperature. Discard the flow-through, and reinsert the spin cartridge into the same collection tube.
5. Repeat Steps 3–4 until the entire sample has been processed.

6. Add 350 μ L Wash Buffer I to the spin cartridge.
7. Centrifuge at $12,000 \times g$ for 30 seconds at room temperature. Discard the flow-through.
8. Prepare DNaseI (10uL DNase I + 70 uL water, 80uL in total per rxn). Add 80 uL DNase I into the column and incubate for 15 min at RT.
9. Add 350 Wash Buffer I to the spin cartridge and centrifuge at $12,000 \times g$ for 30 seconds at room temperature. Discard the flow-through and the collection tube. Place the spin cartridge into a new collection tube.
10. Add 500 μ L Wash Buffer II with ethanol to the spin cartridge.
11. Centrifuge at $12,000 \times g$ for 30 seconds at room temperature. Discard the flow-through.
12. Repeat Steps 8–9 once.
13. Centrifuge the spin cartridge at $12,000 \times g$ for 1–2 minutes to dry the membrane with bound RNA. Discard the collection tube and insert the spin cartridge into a recovery tube.
14. Add 30–100 μ L (50uL) RNase–free water to the center of the spin cartridge.
15. Incubate at room temperature for 1 minute.
16. Centrifuge the spin cartridge for 2 minutes at $\geq 12,000 \times g$ at room temperature to elute the RNA from the membrane into the recovery tube. Note: If the expected RNA yield is $>100 \mu\text{g}$, perform 3 sequential elutions of 100 μ L each. Collect the eluates in a single tube.
17. Store your purified RNA in -80 freezer.

RT with Thermo Scientific Maxima™ H Minus cDNA Synthesis Master Mix with ~~dsDNase~~ (we ran out of dsDNase)

1. Add the following reagents into a sterile, RNase free tube on ice

 image.png

Component	Volume
Maxima H Minus cDNA Synthesis Master Mix (5X)	4 µL
Template RNA	
Total RNA or poly(A) mRNA or Specific RNA	1 pg to 5 µg 0.1 pg to 500 ng 0.01 pg to 500 ng
Water, nuclease-free	to 20 µL
Total volume	20 µL

Table7

	A	B	C	D	E	F	G	H	I	J
1	#	batch	Sample	RNA conc (ng/ul)	A260/280	A260/230	RNA amount for 16uL	minimum amount of RNA	volume for minimum RNA	Water up to 12 uL
2	1	240603EB	4517 hypo 1h	13.8	2.15	0.18	220.8	220.8	16.0	0.0
3	2		5319 2.5mM	15.3	2.19	0.36	244.8	220.8	14.4	1.6
4	3		5319 hypo 1h	15.8	2.36	0.06	252.8	220.8	14.0	2.0
5	4		5319 hypo 24h	17	2.2	1.05	272	220.8	13.0	3.0
6	5	241021EB	BP1h	38.2	2.11	1.42	611.2	446.4	11.7	4.3
7	6		hypo1h	42.3	2.11	0.8	676.8	446.4	10.6	5.4
8	7		hypo24h	30.6	2.13	1.47	489.6	446.4	14.6	1.4
9	8		BP1h + 24R	39.6	2.11	1.87	633.6	446.4	11.3	4.7
10	9		hypo1h + 24R	33	2.11	1.81	528	446.4	13.5	2.5
11	10		hypo24h + 24R	27.9	2.21	0.08	446.4	446.4	16.0	0.0
12	11	241028EB	BP1h	18.9	2.08	1.54	302.4	193.6	10.2	5.8
13	12		hypo1h	14.8	2.16	0.68	236.8	193.6	13.1	2.9
14	13		hypo24h	16.4	2.2	0.22	262.4	193.6	11.8	4.2
15	14		BP1h + 24R	12.1	2.29	0.07	193.6	193.6	16.0	0.0
16	15		hypo1h + 24R	16.3	2.04	1.52	260.8	193.6	11.9	4.1
17	16		hypo24h + 24R	17.2	2.14	0.09	275.2	193.6	11.3	4.7

2. Mix gently and centrifuge.
3. Set up PCR program:

Table11 ^

	A	B
1	temp	time
2	25	10
3	50	15
4	85	5

- a. Incubate at 25 °C for 10 minutes.

b. Incubate at 50 °C for 15 minutes. Note. If using >1 µg RNA template, increase the reaction time to 30 minutes.
For RNA templates that are GC-rich or have a large amount of secondary structure, the reaction temperature can be increased to 65 °C.

c. Terminate the reaction by heating at 85 °C for 5 minutes.
4. The reaction product of the first strand cDNA synthesis can be used directly in qPCR. Store at –20°C for up to one week, or –70°C for long-term storage. Avoid freeze-thaw cycles of the cDNA.

2025年2月19日 星期三

qPCR

Dilute the cDNA 1:5 with nuclease-free water, to have 120 ul/sample total. You will use 3 ul of DNA/qPCR reaction.
Prepare the table to guide the loading of the qPCR.
Use this as template:

241021 EB
wt-11 BP
wt-11 1h Hypo
wt-11 24h Hypo
wt-11 BP+24R
wt-11 1h Hypo +24R
wt-11 24h Hypo +24R

241028EB
wt-11 BP
wt-11 1h Hypo
wt-11 24h Hypo
wt-11 BP+24R
wt-11 1h Hypo +24R
wt-11 24h Hypo +24R

384 well plate																									
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	
1		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
2	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	
3	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	
4	STM2	STM2	STM2	STM2	STM2	STM2	STM2	STM2	STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	STM2	STM2	STM2	STM2	STM2	STM2	
5	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	PFKP	ay_PFKP	ACTB							GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	
6	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	STM2	ay_STM2								ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	
7	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ELAV3	ELAV3	ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	
8	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	PFKP	PFKP	PFKP	ay_PFKP	ay_PFKP	ay_PFKP	
9	STM2	STM2	STM2	STM2	STM2	STM2	STM2	STM2	STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	STM2	STM2	STM2	ay_STM2	ay_STM2	ay_STM2	
10	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH										GAPDH	GAPDH	GAPDH				
11	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB										ACTB	ACTB	ACTB				
12																			ay_ELAV3	ay_PFKP	ay_PFKP	ay_ELAV3	ay_PFKP	ay_PFKP	
13	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_ELAV3	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	
14	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_PFKP	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	
15	STM2	STM2	STM2	STM2	STM2	STM2	STM2	STM2	STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2	ay_STM2							
16	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH																
17	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB																

Enter the number of samples in the yellow cell, number of genes in the green one, and prepare the qPCR mixes accordingly as follows:

Table12						^
	A	B	C	D	E	
1	Reagent	Volume (ul) x1	Volume (uL) (triplicate of all samples +extra 3 wells)	Number of samples	Number of genes to look at	
2	Water, PCR grade (vial 2, colorless cap)	5	195	12	8	
3	PCR Primer mix 10x (5uM)	2	78	Total ul needed (MM)		
4	Master Mix, 2x conc. (vial 1, green cap)	10	390	3120		
5	Total volume	17	663			
6						
7						
8						
9						

- QPCR materials:
- 1. Greiner Sapphire PCR comsumable (384 Well PCR Microplates)
 - 2. Film for 384 well plate (not aluminum or breathwasly film)
 - 3. 8-channel auto-multichannel pipette (Boeke lab)
 - 4. Reagents (RT-PCR box in -20A)
 - a. 2x master mix
 - b. nuclease-free water
 - 5. Tips for multichannel (small, thin ones)
 - 6. 25mL resevoir
 - 7. primers (1:10) from the stock
(primer mixes are in the freezer in TC under bench)

Procedure:

1. setup tip spacing of 8-channel auto-multichannel pipette to 4.7 for 384 wells
2. load DNA with repeat dispense setting of 8-channel auto-multichannel pipette
 - a. dispense: 3uL
 - b. first/last: 3uL
 - c. count: 14
 - d. save
 - e. run for every dispense
 - f. load until last one
 - g. purge run
3. add 17uL master mix for each gene with 8-channel (or maybe 12-well one) auto-multichannel pipette
 - a. setting: pipette -> edit -> adjust volume
4. Run the default on LC480
 - a. make sure the computer of **Felix&LC480** computer is on
 - b. press the corresponding bottom of the **Felix&LC480** computer (e.g., 4)
 - c. log into NYUMC account
 - d. Open LightCycler480 software
 - e. Login:
 - Username: admin
 - Password: LightCycler480
 - f. Put plate in machine by pressing the right button
 - g. Run from template like SYBR Green I
 - detection format: SYBR Green I/HRM Dye
 - check block size: 384
 - volume: 20
 - h. Save experiment
 - i. Start run
5. Export table from LightCycler480
 - a. click Analysis Tab
 - b. create new analysis: Abs Quant/Fit Points or Abs Quant/**2nd Derivative Max**
 - c. select all sample
 - d. click Calculate button at the bottom to calculate Cp number
 - e. Export the table as .txt

Analyze the experiment. You can use [this table](#) as a template.

2025年10月6日 星期一

Updated Protocol!!!

RNA Samples:

old sample (may not use)				^
	A	B	C	
1		differentiation	experiment	
2	20250310MN	BP T8 w/ 1.5pyruvate	NBA without sodium pyruvate	
3	20250407MN	2.5mM T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A)	2.5mM T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A)	
4	20250414MN	2.5mM T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A)	2.5mM T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A)	
5	20250505MN	2.5mM T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A)	2.5mM T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A)	

sample with the most uptodate protocol						^
	A	B	C	D	E	
1		differentiation	experiment	sample#	from what well plate	
2	20250630MN	BP T8 w/ 1.5pyruvate	NBA w/ 1mM pyruvate	6	6	
3	20250728MN	BP T8 w/ 1.5pyruvate	NBA w/ 1mM pyruvate	6	6	
4	20250908MN	BP T8 w/ 1.5pyruvate	NBA w/ 1mM pyruvate	4	6	

RNA extraction with TRIzol (protocol)

Lyse samples and separate phases

1. Lyse and homogenize samples in TRIzol™ Reagent
2. Cell grown in monolayer:
 - a. Remove growth media.
 - b. Add 1 mL of TRIzol™ Reagent per 1×10^5 — 1×10^7 cells directly to the 3.5-cm culture dish to lyse the cells.
(The trizol froze when I added it to the plate which I stored in -80)
 - c. Pipet the lysate up and down several times to homogenize (I put the plate on the bench in RT, pipetted it, and let it thaw on the plate)
3. (Optional, I didn't do) If samples have a high fat content, centrifuge the lysate for 5 minutes at $12,000 \times g$ at $4-10^\circ\text{C}$, then transfer the clear supernatant to a new tube. 3. Incubate for 5 minutes to allow complete dissociation of the nucleoproteins complex.

===I froze samples at this step. I took them out today and thawed them on the bench.=====

1. Add 0.2 mL of chloroform per 1 mL of TRIzol™ Reagent used for lysis, securely cap the tube, then thoroughly mix by shaking.
2. Incubate for 2–3 minutes.
3. Centrifuge the sample for 15 minutes at 12,000 × g at 4°C. The mixture separates into a lower phenol-chloroform, an interphase, and a colorless upper aqueous phase.
4. Transfer the aqueous phase containing the RNA to a new tube by angling the tube at 45° and pipetting the solution out. (can only get 500uL/tube)
5. Precipitate the RNA
 - a. (Optional, I didn't do) If the starting sample is small (<106 cells or <10 mg of tissue), add 5–10 µg of RNase-free glycogen as a carrier to the aqueous phase. Note: The glycogen is co-precipitated with the RNA, but does not interfere with subsequent applications.
 - b. Add 0.5 mL of isopropanol (1 volume) to the aqueous phase, per 1 mL of TRIzol™ Reagent used for lysis.
 - c. Incubate for 10 minutes at 4°C. (incubated in -20 o/n instead to increase the yield)

2025年10月7日 星期二

- a. Centrifuge for 10 minutes (did 30 min instead) at 12,000 × g at 4°C. Total RNA precipitate forms a white gel-like pellet at the bottom of the tube.
 - b. Discard the supernatant with a micropipettor.
1. Wash the RNA
 - a. Resuspend the pellet in 1 mL of 75% ethanol per 1 mL of TRIzol™ Reagent used for lysis. Note: The RNA can be stored in 75% ethanol for at least 1 year at –20°C, or at least 1 week at 4°C.
 - b. Vortex the sample briefly, then centrifuge for 5 minutes at 7500 × g at 4°C.
 - c. Discard the supernatant with a micropipettor. (I did two 10min 12000xg at 4C wash and one 5min 12000xg 4C after the second wash. I found the RNA moved easily, so I used 10uL pipette to try to remove the extra ethanol (I can't do much since the RNA move with ethanol).)
 - d. Vacuum or air dry the RNA pellet for 5–10 minutes. (I did air dry with tubes hanged on the racks. a bit more than 30 minutes in total) IMPORTANT! Do not dry the pellet by vacuum centrifuge. Do not let the RNA pellet dry, to ensure total solubilization of the RNA. Partially dissolved RNA samples have an A230/280 ratio <1.6.
2. Solubilize the RNA
 - a. Resuspend the pellet in 20–50 µL (I used 18uL) of RNase-free water, 0.1 mM EDTA, or 0.5% SDS solution by pipetting up and down. IMPORTANT! Do not dissolve the RNA in 0.5% SDS if the RNA is to be used in subsequent enzymatic reactions.
 - b. Incubate in a water bath or heat block set at 55–60°C for 10–15 minutes. Proceed to RT and store the rest of RNA at –80°C.

Note: I might accidentally aspirate the RNA when I tried to remove the ethanol. Should not centrifuge again after removing the ethanol. Let the RNA dry longer time.

Table4											
	A	B	C	D	E	F	G	H	I	J	K
1	#	batch	Sample		RNA conc (ng/ul)	A260/280	A260/230	RNA amount for 10uL (ng)	minimum amount of RNA	volume for minimum RNA	Water up to 12 uL
2	1	250630MN	A1	2.5mm + DMSO 6h	4	1.54	0.03	40	986	10.0	0.0
3	2		A2	2.5mM glu 1h	215	1.88	0.79	2150	986	4.6	5.4
4	3		A3	2.5mM glu 24h	98.6	1.71	1.9	986	986	10.0	0.0
5	4		B1	2.5mM + 2uM MG132 6h	322.1	1.81	2.09	3221	986	3.1	6.9
6	5		B2	0.15mM glu 1h	338.7	1.65	2.15	3387	986	2.9	7.1
7	6		B3	0.15mM glu 24h	369.4	1.71	0.82	3694	986	2.7	7.3
8	7	250728MN	A1	2.5mM glucose 1h	23.1	1.8	0.12	231	951	10.0	0.0
9	8		A2	2.5mM glucose 24h	163.9	1.82	1.29	1639	951	5.8	4.2
10	9		A3	2.5mM glucose +DMSO 24h	204.2	1.8	1.69	2042	951	4.7	5.3
11	10		B1	0.15mM glucose 1h	200.4	1.91	1.22	2004	951	4.7	5.3
12	11		B2	0.15mM glucose 24h	152.2	1.85	0.75	1522	951	6.2	3.8
13	12		B3	2.5mM glucose +2uM MG132 24h	95.1	1.7	2.17	951	951	10.0	0.0
14	13	250908MN	A1	2.5mM gluc 1h	101	1.57	0.66	1010	1010	10.0	0.0
15	14		A2	2.5mM gluc 24h	118	1.53	0.63	1180	1010	8.6	1.4
16	15		B1	0.15mM gluc 1h	1.6	1.33	0.85	16	1010	10.0	0.0
17	16		B2	0.15mm gluc 24h	117.8	1.88	0.62	1178	1010	8.6	1.4

RT with Thermo Scientific Maxima™ H Minus cDNA Synthesis Master Mix with dsDNase

Table1			
	A	B	C
1	sample#	16	
2	10X dsDNase Buffer	1 µL	17
3	dsDNase	1 µL	17
4	Template RNA •Total RNA or •poly(A) mRNA or •Specific RNA	1 pg to 5 µg 0.1 pg to 500 ng 0.01 pg to 500 ng	
5	Water, nuclease-free	to 12 µL	
6	Total volume	12 µL	

1. Prepare mix of
 - a. 1ul DNase/sample +1
 - b. 1ul of buffer 10x/sample +1
2. Add 2 ul/ tube. (add 2.2uL this time)
3. Mix gently and centrifuge.
4. Incubate at 37°C in a preheated thermomixer or water bath for 2 minutes.
5. Chill on ice, briefly centrifuge and place on ice.
6. Add the following components to the tube on ice:

Table2			
	A	B	C
1	Maxima cDNA H Minus Synthesis Master Mix (5X)	4 µL	
2	Water, nuclease-free	4 µL	

1. Mix gently and centrifuge.
2. Set up PCR program:

Table6 ^

	A	B
1	temp	time
2	25	10
3	50	15
4	85	5

- a. Incubate at 25 °C for 10 minutes.
- b. Incubate at 50 °C for 15 minutes. Note. If using >1 µg RNA template, increase the reaction time to 30 minutes. For RNA templates that are GC-rich or have a large amount of secondary structure, the reaction temperature can be increased to 65 °C.
- c. Terminate the reaction by heating at 85 °C for 5 minutes.

***The reaction product of the first-strand cDNA synthesis can be used directly in qPCR. Store at –20°C for up to one week or –70°C for long-term storage. Avoid freeze-thaw cycles of the cDNA.

Dilute the cDNA 1:5 with nuclease-free water, to have 120 ul/sample total. -> store inn -20C freezer

qPCR

use 3 ul of DNA/qPCR reaction.
Prepare the table to guide the loading of the qPCR.
Use this as template:

genes: STM2, STM2-cry, ACTB, GAPDH

Well3																								
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
A	1	1	1	2	2	2	3	3	3	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8
B	1	1	1	2	2	2	3	3	3	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8
C	1	1	1	2	2	2	3	3	3	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8
D	1	1	1	2	2	2	3	3	3	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8
E																								
F	9	9	9	10	10	10	11	11	11	12	12	12	13	13	13	14	14	14	15	15	15	16	16	16
G	9	9	9	10	10	10	11	11	11	12	12	12	13	13	13	14	14	14	15	15	15	16	16	16
H	9	9	9	10	10	10	11	11	11	12	12	12	13	13	13	14	14	14	15	15	15	16	16	16
I	9	9	9	10	10	10	11	11	11	12	12	12	13	13	13	14	14	14	15	15	15	16	16	16
J																								
K																								
L																								
M																								
N																								
O																								
P																								

Enter the number of samples in the yellow cell, number of genes in the green one, and prepare the qPCR mixes accordingly as follows:

Table5					
	A	B	C	D	E
1	Reagent	Volume (ul) x1	Volume (uL) (triplicate of all samples +extra 3 wells)	Number of samples	Number of genes to look at
2	Water, PCR grade (vial 2, colorless cap)	5	255	16	4
3	PCR Primer mix 10x (5uM)	2	102	Total ul needed (MM)	
4	Master Mix, 2x conc. (vial 1, green cap)	10	510	2040	
5	Total volume	17	867		

QPCR materials:

- a. Greiner Sapphire PCR consumable (384 Well PCR Microplates)
- b. Film for 384 well plate (not aluminum or breathwasy film)
- c. 8-channel (12-channel) auto-multichannel pipette (Boeke lab)
- d. Reagents (RT-PCR box in -20A)
 - I. 2x master mix

- II. nuclease-free water
- e. Tips for multichannel (small, thin ones)
- f. 25mL resevoir
- g. primers (1:10) from the stock
(primer mixes are in the freezer in TC under bench)

Procedure:

- a. setup tip spacing of 8-channel auto-multichannel pipette to 4.7 for 384 wells
- b. load DNA with repeat dispense setting of 8-channel auto-multichannel pipette
 - I. dispense: 3uL
 - II. first/last: 3uL
 - III. count: 14
 - IV. save
 - V. run for every dispense
 - VI. load until last one
 - VII. purge run
- c. add 17uL master mix for each gene with 8-channel (or maybe 12-well one) auto-multichannel pipette
 - I. setting: pipette -> edit -> adjust volume
- d. Run the default on LC480
 - I. make sure the computer of **Felix&LC480** computer is on
 - II. press the corresponding bottom of the **Felix&LC480** computer (e.g., **4**)
 - III. log into NYUMC account (Boeke lab account)
 - IV. Open LightCycler480 software
 - V. Login:
 - Username: admin
 - Password: LightCycler480
 - VI. Put plate in machine by pressing the right botton
 - VII. Run from template like SYBR Green I
 - detection format: SYBR Green I/HRM Dye
 - check block size: 384
 - volume: 20
 - VIII. Save experiment
 - IX. Start run
- e. Export table from LightCycler480
 - I. click Analysis Tab
 - II. create new analysis: Abs Quant/Fit Points or Abs Quant/**2nd Derivative Max**
 - III. select all sample
 - IV. click Calculate botton at the bottom to calculate Cp number
 - V. Export the table as .txt

Analyze the experiment. You can use [this table](#) as a template.

****NOTE:** This protocol uses too much reagents. Try to cut in a half (10uL in total) next time. Chao said he used 7uL in total and it worked well. The RT step can also reduced to 10uL in total.

2025年12月24日 星期三

RNA sample:

sample with the most uptodate protocol1								^
	A	B	C	D	E	F	G	
1		differentiation	experiment	sample #	well-plate	collect on	trizol amount	
2	20251103MN	BP T8 w/ 1.5pyruvate	NBA w/ 1mM pyruvate	8	12	20251126	200uL	

RNA extraction with Direct-zol™ RNA Miniprep (protocol)

1. Lyse the samples with 200uL Tri Reagent
- ===I froze samples at this step. I took them out today and thawed them on the bench.=====
- RNA Purification
- ✓ Perform all steps at room temperature and centrifugation at 10,000-16,000 x g for 30 seconds, unless specified.
1. Add an equal volume ethanol (95-100%) to a sample lysed in TRI Reagent® or similar1 and mix thoroughly.
- Example: Add 400 µl ethanol to 400 µl mixture (sample lysed in TRI Reagent®).
2. Transfer the mixture into a Zymo-Spin™ IICR Column2 in a Collection Tube and centrifuge. Transfer the column into a new collection tube and discard the flow-through.
3. DNase I4 treatment (recommended) I did this
- (D1) Add 400 µl RNA Wash Buffer to the column and centrifuge.
- (D2) In an RNase-free tube (not included), add 5 µl DNase I (6 U/µl)*, 75 µl DNA Digestion Buffer and mix by gentle inversion. Add the mix directly to the column matrix.
- The DNase I we have is 2000U/mL --> 30U = 15uL

Table3

	A	B	C	D
1	sample	8		
2		volume	total	actual total
3	DNase I 30U	5	40	42
4	DNA Digestion Buffer	75	600	630



(D3) Incubate at room temperature (20-30°C) for 15 minutes. Proceed to step 4.

4. Add 400 µl Direct-zol™ RNA PreWash5 to the column and centrifuge. Discard the flow-through and repeat this step.
5. Add 700 µl RNA Wash Buffer5 to the column and centrifuge for 1 minute to ensure complete removal of the wash buffer. Transfer the column carefully into an RNase-free tube (not included).
6. To elute RNA, add 50 µl of DNase/RNase-Free Water directly to the column matrix and centrifuge. **(Use 30uL)**
Alternatively, for highly concentrated RNA use ≥ 25 µl elution

Sample list

	A	B	C	D	E	F	G
1	#	batch	Sample		RNA conc (ng/ul)	A260/280	A260/230
2	1	251103MN	A1	2.5mM glu 1h	1.3	1.56	0.98
3	2		A2	2.5mM glu 1h	0.9	1.12	0.38
4	3		A3	2.5mM glu 24h	1.4	1.28	0.66
5	4		A4	2.5mM glu 24h	0.7	0.95	0.46
6	5		B1	0.15mM glu 1h	1.3	0.94	0.64
7	6		B2	0.15mM glu 1h	1.6	1.05	0.8
8	7		B3	0.15mM glu 24h	1.5	1.71	1.27
9	8		B4	0.15mM glu 24h	1.8	1.14	0.1



RNA concentrations were way too low --> use 5uL of RNA for all.

Since we already treated the DNaseI, I will proceed to the RT directly

RT

1. Add the following components to the tube on ice:

Table9				^
	A	B	C	
1	sample	8		
2		saving volume (uL)	total	
3	Maxima cDNA H Minus Synthesis Master Mix (5X)	2	17	
4	Water, nuclease-free	3	25.5	

2. Prepare mix of
- 2ul Maxima cDNA H Minus Synthesis Master Mix (5X)/sample +1
 - 3ul of Water, nuclease-free/sample +1
3. Add 5 ul MM per tube.
4. Add 5uL RNA (or RNA + H2O)
5. Mix gently and centrifuge. Load the program in the PCR machine:

Table10			^
	A	B	
1	temp	time	
2	25	10	
3	50	15	
4	85	5	

- Incubate at 25°C for 10 minutes.
- Incubate at 50°C for 15 minutes.
- Terminate the reaction by heating at 85°C for 5 minutes.
6. Use directly or freeze in -80 for long term storage.

qPCR ("Saving" protocol)

- 1. Dilute the cDNA 1:3 with nuclease-free water. Add 2uL DNA/qPCR reaction.
- 2. Prepare the table to guide the loading of the qPCR.

genes: STM2, STM2-cry, ACTB, GAPDH

Well1																								
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
A	2	2	2	3	3	3	1	1	1	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8
B	2	2	2	3	3	3	1	1	1	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8
C	2	2	2	3	3	3	1	1	1	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8
D	2	2	2	3	3	3	1	1	1	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8
E																								
F	9	9	9	10	10	10	11	11	11	12	12	12	13	13	13	14	14	14	15	15	15	16	16	16
G	9	9	9	10	10	10	11	11	11	12	12	12	13	13	13	14	14	14	15	15	15	16	16	16
H	9	9	9	10	10	10	11	11	11	12	12	12	13	13	13	14	14	14	15	15	15	16	16	16
I	9	9	9	10	10	10	11	11	11	12	12	12	13	13	13	14	14	14	15	15	15	16	16	16
J																								
K																								
L																								
M																								
N																								
O																								
P																								

Enter the number of samples in the yellow cell, number of genes in the green one, and prepare the qPCR mixes accordingly as follows:

Table8					
	A	B	C	D	E
1		10uL			
2	Reagent	Volume (ul) x1	Volume (uL) (triplicate of all samples +extra 3 wells)	Number of samples	Number of genes to look at
3	Water, PCR grade (vial 2, colorless cap)	2.5	67.5	8	4
4	PCR Primer mix 10x (5uM)	1	27	Total ul needed (MM)	Total ul needed (H2O)
5	Master Mix, 2x conc. (vial 1, green cap)	5	135	540	270
6	Total volume	8.5	229.5		

QPCR materials:

- a. Greiner Sapphire PCR comsumable (384 Well PCR Microplates)

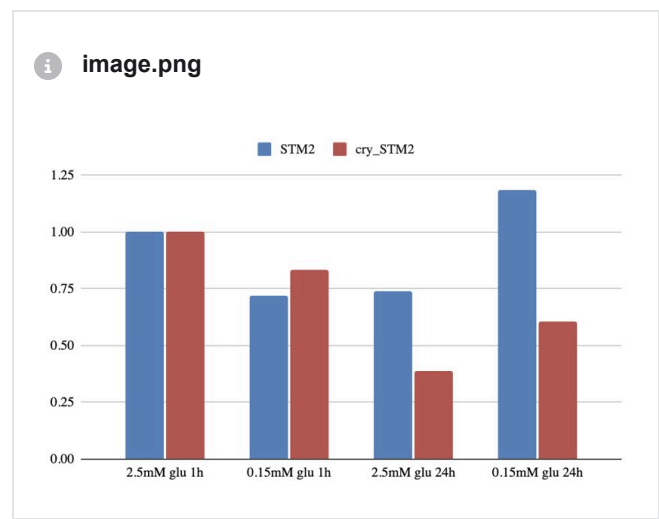
- b. Film for 384 well plate (not aluminum or breathwasy film)
- c. 8-channel (12-channel) auto-multichannel pipette (Boeke lab)
- d. Reagents (RT-PCR box in -20A)
 - I. 2x master mix
 - II. nuclease-free water
- e. Tips for multichannel (small, thin ones)
- f. 25mL resevoir
- g. primers (1:10) from the stock
(primer mixes are in the freezer in TC under bench)

Procedure:

- a. load DNA with pipette setting of 8-channel auto-multichannel pipette from 8-tube strip
 - I. set up tip spacing: 3 steps -> 4.5mm, 9mm, 13.5mm
 - II. pipette volume: 2uL
 - III. save
 - IV. Aspirate with 9mm and dispense with 13.5mm (it's three replicate/sample). Change tips every row.
- b. transfer primer master mix into 4*8-tube strip.
- c. Add 8.5 uL master mix for each gene. Aspirate with 9mm and dispense with 4.5mm by column. (with this way, I don't need resevoir and it will have more extra volume (perhaps I can prepare less extra reagent)
 - I. setting: pipette -> edit -> adjust volume
- d. Run the default on LC480
 - I. make sure the computer of **Felix&LC480** computer is on
 - II. press the corresponding bottom of the **Felix&LC480** computer (e.g., 4)
 - III. log into NYUMC account (Boeke lab account)
 - IV. Open LightCycler480 software
 - V. Login:
 - Username: admin
 - Password: LightCycler480
 - VI. Put plate in machine by pressing the right botton
 - VII. Run from template like SYBR Green I
 - detection format: SYBR Green I/HRM Dye
 - check block size: 384
 - volume: 20
 - VIII. Save experiment
 - IX. Start run
- e. Export table from LightCycler480
 - I. click Analysis Tab
 - II. create new analysis: Abs Quant/Fit Points or Abs Quant/**2nd Derivative Max**
 - III. select all sample
 - IV. click Calculate botton at the bottom to calculate Cp number
 - V. Export the table as .txt

Analyze the experiment with this [spreadsheet](#)

The Cp number of crySTMN2 was very low. Some sample was not detectable
It might because the RNA concentration was very low....
May want to use traditional way to extract RNA next time



2026年4月7日 星期二

Thaw the following smaple

sample with the most uptodate protocol2						
	A	B	C	D	E	F
1		differentiation	experiment	sample#	from what kind of well plate	Where did I store Trizol samples
2	20260224MN	BP T8 w/ additional 1mM pyruvate (1.5mM in total)	NBA w/ 1mM pyruvate	4	6	I froze in -80 freezer
3	20260302MN	BP T8 w/ additional 1mM pyruvate (1.5mM in total)	NBA w/ 1mM pyruvate	6	6	I froze in -20 freezer
4	20260309MN	BP T8 w/ additional 1mM pyruvate (1.5mM in total)	NBA w/o pyruvate	6	6	I froze in -80 freezer

RNA extraction with TRIzol (protocol)

- Lyse samples and separate phases
- 1. Lyse and homogenize samples in TRIzol™ Reagent
 - 2. Cell grown in monolayer:
 - a. Remove growth media.
 - b. Add 1 mL of TRIzol™ Reagent per 1 × 105—1 × 107 cells directly to the 3.5–cm culture dish to lyse the cells.
(The trizol froze when I added it to the plate which I stored in -80)
 - c. Pipet the lysate up and down several times to homogenize (I put the plate on the bench in RT, pipetted it, and let it thaw on the plate)

3. (Optional, I didn't do) If samples have a high fat content, centrifuge the lysate for 5 minutes at $12,000 \times g$ at $4-10^{\circ}\text{C}$, then transfer the clear supernatant to a new tube. 3. Incubate for 5 minutes to allow complete dissociation of the nucleoproteins complex.

==I froze samples at this step. I took them out today and thawed them on the bench today.=====

1. Add 0.2 mL of chloroform per 1 mL of TRIzol™ Reagent used for lysis, securely cap the tube, then thoroughly mix by shaking.
2. Incubate for 2–3 minutes.
3. Centrifuge the sample for 15 minutes at $12,000 \times g$ at 4°C . The mixture separates into a lower phenol-chloroform, an interphase, and a colorless upper aqueous phase.
4. Transfer the aqueous phase containing the RNA to a new tube by angling the tube at 45° and pipetting the solution out. (can only get around 500uL/tube)
5. Precipitate the RNA
 - a. (Optional, I didn't do) If the starting sample is small (<106 cells or <10 mg of tissue), add 5–10 μg of RNase-free glycogen as a carrier to the aqueous phase. Note: The glycogen is co-precipitated with the RNA, but does not interfere with subsequent applications.
 - b. Add 0.5 mL of isopropanol (1 volume) to the aqueous phase, per 1 mL of TRIzol™ Reagent used for lysis.
 - c. Incubate for 10 minutes at 4°C . (incubated in -20°C instead to increase the yield)

2026年4月8日 星期三

- a. Centrifuge for 10 minutes (did 30 min instead) at $12,000 \times g$ at 4°C . Total RNA precipitate forms a white gel-like pellet at the bottom of the tube. (I didn't see any pellet at this point. The last time I did the Trizol protocol, the pellets were pretty clear)
- b. Discard the supernatant with a micropipettor.
1. Wash the RNA
 - a. Resuspend the pellet in 1 mL of 75% ethanol per 1 mL of TRIzol™ Reagent used for lysis. Note: The RNA can be stored in 75% ethanol for at least 1 year at -20°C , or at least 1 week at 4°C .
 - b. Vortex the sample briefly, then centrifuge for 5 minutes at $7500 \times g$ at 4°C .
 - c. Discard the supernatant with a micropipettor. (I did two 10min $12000 \times g$ at 4°C wash and one 1min $12000 \times g$ 4°C after the second wash. Most samples did not have visible pellet, I tried to remove the top ethanol with P1000 and leave a bit at the bottom for the first two rounds. After 1min final centrifuge, I use P100 to remove the residual gently at the opposite side of the tube. I hope I save the RNA...)
 - d. Vacuum or air dry the RNA pellet for 5–10 minutes. (After I removed the residual ethanol, I opened the lid and let the RNA air dry on the bench.)
 - e. IMPORTANT! Do not dry the pellet by vacuum centrifuge. Do not let the RNA pellet dry, to ensure total solubilization of the RNA. Partially dissolved RNA samples have an A_{230}/A_{280} ratio <1.6 .
2. Solubilize the RNA
 - a. Resuspend the pellet in 20–50 μL (I used 18uL) of RNase-free water, 0.1 mM EDTA, or 0.5% SDS solution by pipetting up and down. IMPORTANT! Do not dissolve the RNA in 0.5% SDS if the RNA is to be used in subsequent enzymatic reactions.
 - b. Incubate in a water bath or heat block set at $55-60^{\circ}\text{C}$ for 10–15 minutes. Proceed to RT and store the rest of RNA at -80°C .

Table13

	A	B	C	D	E	F	G	H
1	#	batch	Sample		replicate	RNA conc (ng/ul)	A260/280	A260/230
2	1	20260224MN	A1	Ctrl	A	57.2	1.7	1.65
3	2		A2	RLG	A	41.7	1.69	1.85
4	3		B1	Ctrl	B	38.5	1.7	1.56
5	4		B2	RLG	B	61.2	1.69	1.85
6	5	20260302MN	A1	2.5mM gluc 1h	A	81.5	1.67	1.9
7	6		A2	2.5mM gluc 1h	A	84.1	1.64	2.1
8	7		A3	0.15mM gluc 1h	A	115.3	1.71	2.1
9	8		B1	2.5mM gluc 1h	B	59.7	1.61	2.04
10	9		B2	0.15mM gluc 1h	B	63.3	1.61	2.02
11	10		B3	0.15mM gluc 1h	B	69.2	1.63	1.99
12	11	20260309MN	A1	2.5mM gluc w/o pyruvate 24h	A	126	1.7	2.14
13	12		A2	2.5mM gluc w/o pyruvate 1h	A	143.2	1.76	1.95
14	13		A3	2.5mM gluc w/o pyruvate 1h	B	110.8	1.75	2.03
15	14		B1	0.15mM gluc w/o pyruvate 24h	A	96	1.69	2.05
16	15		B2	0.15mM gluc w/o pyruvate 1h	A	85.5	1.73	1.92
17	16		B3	0.15mM gluc w/o pyruvate 1h	B	109.4	1.74	1.88

RT

DNase treatment in PCR tube. If not process with dsDase, use 5uL RNA template and do step 14 directly.

Table17

	A	B	C	D
1	sample #	16		
2		standard	saving+ (use maximum RNA)	
3	10X dsDNase Buffer	1 µL	0.8	13.6
4	dsDNase	1 µL	0.8	13.6
5	Template RNA •Total RNA or •poly(A) mRNA or •Specific RNA	1 pg to 5 µg 0.1 pg to 500 ng 0.01 pg to 500 ng	8uL	
6	Water, nuclease-free	to 12 µL	to 8uL	
7	Total volume	12 µL	8uL	

1. Prepare mix of
- a. 0.5ul DNase/sample +1

b. 0.5ul of buffer 10x/sample +1

c. Add 1 ul/ tube.
2. Mix gently and centrifuge.
3. Incubate at 37°C in a preheated thermomixer or water bath for 2 minutes.
4. Chill on ice, briefly centrifuge and place on ice.

Add the following components to the tube on ice:

Table18

	A	B	C	D
1	sample#	16		
2		standar d	saving	
3	Maxima cDNA H Minus Synthesis Master Mix (5X)	4 µL	2uL	
4	Water, nuclease-free	4 µL	0uL	

- Prepare mix of (add 2uL Master Mix to sample)
- 2uL Maxima cDNA H Minus Synthesis Master Mix (5X)/sample
 - ~~3ul of Water, nuclease-free/sample +1~~
 - ~~Add 5 ul per tube.~~

Mix gently and centrifuge. Load the program in the PCR machine:

Table19			^
	A	B	
1	temp	time	
2	25	10	
3	50	15	
4	85	5	

- Incubate at 25°C for 10 minutes.
- Incubate at 50°C for 15 minutes.
- Terminate the reaction by heating at 85°C for 5 minutes.
- > freeze in -20.

2026年5月21日 星期四

Thaw the following smaple

sample with the most uptodate protocol3									✓
	A	B	C	D	E	F	G	H	
1		differentiation	experiment	sample #	from what kind of well plate	Where did I store Trizol samples	note1	note2	
2	20260415MN	BP T8 w/ additional 1mM pyruvate (1.5mM in total)	NBA w/ 1mM pyruvate	6	12	I froze in -80 freezer	4 of them were combined from 2 12 wells	MMS might be hydrolyzed at this point since I prepared the media around noon.	
3			2.5mM gluc	6	6	I froze in -80 freezer			
4	20260420MN	BP T8 w/ additional 1mM pyruvate (1.5mM in total)	NBA w/ 1mM pyruvate	3	12	I froze in -80 freezer			

RNA extraction with TRIZol (protocol)

- Lyse samples and separate phases
1. Lyse and homogenize samples in TRIZol™ Reagent
 2. Cell grown in monolayer:
 - a. Remove growth media.
 - b. Add 1 mL of TRIZol™ Reagent per 1 × 105—1 × 107 cells directly to the 3.5–cm culture dish to lyse the cells.
(The trizol froze when I added it to the plate which I stored in -80)
 - c. Pipet the lysate up and down several times to homogenize (I put the plate on the bench in RT, pipetted it, and let it thaw on the plate)
 3. (Optional, I didn't do) If samples have a high fat content, centrifuge the lysate for 5 minutes at 12,000 × g at 4–10°C, then transfer the clear supernatant to a new tube. 3. Incubate for 5 minutes to allow complete dissociation of the nucleoproteins complex.

===I froze samples at this step. I took them out today and thawed them on the bench today.=====

1. Add 0.2 mL of chloroform per 1 mL of TRIzol™ Reagent used for lysis, securely cap the tube, then thoroughly mix by shaking.
2. Incubate for 2–3 minutes.
3. Centrifuge the sample for 15 minutes at 12,000 × g at 4°C. The mixture separates into a lower phenol-chloroform, an interphase, and a colorless upper aqueous phase.
4. Transfer the aqueous phase containing the RNA to a new tube by angling the tube at 45° and pipetting the solution out.
(can only get around 500uL/tube)
5. Precipitate the RNA
 - a. (Optional, I didn't do) If the starting sample is small (<106 cells or <10 mg of tissue), add 5–10 µg of RNase-free glycogen as a carrier to the aqueous phase. Note: The glycogen is co-precipitated with the RNA, but does not interfere with subsequent applications.
 - b. Add 0.5 mL of isopropanol (1 volume) to the aqueous phase, per 1 mL of TRIzol™ Reagent used for lysis.
 - c. Incubate for 10 minutes at 4°C. (incubated in -20 o/n instead to increase the yield)

2026年5月22日 星期五

- a. Centrifuge for 10 minutes (did 30 min instead) at 12,000 × g at 4°C. Total RNA precipitate forms a white gel-like pellet at the bottom of the tube. (I didn't see any pellet at this point again. The last time I did the Trizol protocol, the pellets were pretty clear)
- b. Discard the supernatant with a micropipettor. (I didn't use micropipettor this time, I dump the supernatent directly)
1. Wash the RNA
 - a. Resuspend the pellet in 1 mL of 75% ethanol per 1 mL of TRIzol™ Reagent used for lysis. Note: The RNA can be stored in 75% ethanol for at least 1 year at –20°C, or at least 1 week at 4°C.
 - b. Vortex the sample briefly, then centrifuge for 5 minutes at 7500 × g at 4°C.
 - c. Discard the supernatant with a micropipettor. (I did two 10min 12000xg at 4C wash and one 1min 12000xg 4C after the second wash. Most samples did not have visible pellet, I didn't use micropipettor this time, I dump the supernatent directly for the first two rounds. After 1min final centrifuge, I use P100 to remove the residual gently at the opposite side of the tube. I hope I save the RNA...)
 - d. Vacuum or air dry the RNA pellet for 5–10 minutes. (After I removed the residual ethanol, I opened the lid and let the RNA air dry on the bench.)
 - e. IMPORTANT! Do not dry the pellet by vacuum centrifuge. Do not let the RNA pellet dry, to ensure total solubilization of the RNA. Partially dissolved RNA samples have an A230/280 ratio <1.6.
2. Solubilize the RNA
 - a. Resuspend the pellet in 20–50 µL (I used 18uL) of RNase-free water, 0.1 mM EDTA, or 0.5% SDS solution by pipetting up and down. IMPORTANT! Do not dissolve the RNA in 0.5% SDS if the RNA is to be used in subsequent enzymatic reactions.
 - b. Incubate in a water bath or heat block set at 55–60°C for 10–15 minutes. Proceed to RT and store the rest of RNA at –80°C.

Table15								
	A	B	C	D	E	F	G	H
1	#	batch	Sample		cell	RNA conc (ng/ul)	A260/280	A260/230
2	17	20260415MN	A1+A3	0mM MMS	iPSC1000			
3	18		A2+A4	1.5mM MMS	iPSC1000			
4	19		B1+B3	0mM MMS	iPSC1066			
5	20		B2+B4	1.5mM MMS	iPSC1066			
6	21		C1	0mM MMS	iPSC1108			
7	22		C2	1.5mM MMS	iPSC1108			
8	23	20260420MN	A1	0mM MMS	iPSC1000			
9	24		B1	1.5mM MMS	iPSC1000			
10	25		A2	0mM MMS	iPSC1066			
11	26		B2	1.5mM MMS	iPSC1066			
12	27		A3	0mM MMS + 200nM olaparib	iPSC1066			
13	28		B3	1.5mM MMS+ 200nM olaparib	iPSC1066			
14	29		A1	0mM MMS	iPSC1108			
15	30		B1	1.5mM MMS	iPSC1108			
16	31		C1	1.5mM MMS	iPSC1108			

RT

DNase treatment in PCR tube. If not process with dsDase, use 5uL RNA template and do step 14 directly.

Table16

	A	B	C	D
1	sample #	15		
2		standard	saving+ (use maximum RNA)	
3	10X dsDNase Buffer	1 µL	0.8	12.8
4	dsDNase	1 µL	0.8	12.8
5	Template RNA •Total RNA or •poly(A) mRNA or •Specific RNA	1 pg to 5 µg 0.1 pg to 500 ng 0.01 pg to 500 ng	8uL	
6	Water, nuclease-free	to 12 µL	to 8uL	
7	Total volume	12 µL	8uL	

1. Prepare mix of
- a. 0.5ul DNase/sample +1

b. 0.5ul of buffer 10x/sample +1

c. Add 1 ul/ tube.
2. Mix gently and centrifuge.
3. Incubate at 37°C in a preheated thermomixer or water bath for 2 minutes.
4. Chill on ice, briefly centrifuge and place on ice.

Add the following components to the tube on ice:

Table20

	A	B	C	D
1	sample#	15		
2		standar d	saving	
3	Maxima cDNA H Minus Synthesis Master Mix (5X)	4 µL	2.4uL	
4	Water, nuclease-free	4 µL	0uL	

- Prepare mix of (add 2uL Master Mix to sample)
- 2.4ul Maxima cDNA H Minus Synthesis Master Mix (5X)/sample
 - ~~3ul of Water, nuclease-free/sample +1~~
 - ~~Add 5 ul per tube.~~

Mix gently and centrifuge. Load the program in the PCR machine:

Table21			^
	A	B	
1	temp	time	
2	25	10	
3	50	15	
4	85	5	

Incubate at 25°C for 10 minutes.
Incubate at 50°C for 15 minutes.
Terminate the reaction by heating at 85°C for 5 minutes.
-> freeze in -20.

QPCR

qPCR ("Saving" protocol)

1. Dilute the cDNA 1:3 with nuclease-free water. Add 2uL DNA/qPCR reaction.
2. Prepare the table to guide the loading of the qPCR.

genes: STM2, STM2-cry, ACTB, GAPDH

Well2																								
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
A	1	1	1	2	2	2	3	3	3	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8
B	1	1	1	2	2	2	3	3	3	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8
C	1	1	1	2	2	2	3	3	3	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8
D	1	1	1	2	2	2	3	3	3	4	4	4	5	5	5	6	6	6	7	7	7	8	8	8
E	9	9	9	10	10	10	11	11	11	12	12	12	13	13	13	14	14	14	15	15	15	16	16	16
F	9	9	9	10	10	10	11	11	11	12	12	12	13	13	13	14	14	14	15	15	15	16	16	16
G	9	9	9	10	10	10	11	11	11	12	12	12	13	13	13	14	14	14	15	15	15	16	16	16
H	9	9	9	10	10	10	11	11	11	12	12	12	13	13	13	14	14	14	15	15	15	16	16	16
I	17	17	17	18	18	18	19	19	19	20	20	20	21	21	21	22	22	22	23	23	23	24	24	24
J	17	17	17	18	18	18	19	19	19	20	20	20	21	21	21	22	22	22	23	23	23	24	24	24
K	17	17	17	18	18	18	19	19	19	20	20	20	21	21	21	22	22	22	23	23	23	24	24	24
L	17	17	17	18	18	18	19	19	19	20	20	20	21	21	21	22	22	22	23	23	23	24	24	24
M	25	25	25	26	26	26	27	27	27	28	28	28	29	29	29	30	30	30	31	31	31	NC		
N	25	25	25	26	26	26	27	27	27	28	28	28	29	29	29	30	30	30	31	31	31	NC		
O	25	25	25	26	26	26	27	27	27	28	28	28	29	29	29	30	30	30	31	31	31	NC		
P	25	25	25	26	26	26	27	27	27	28	28	28	29	29	29	30	30	30	31	31	31	NC		

Enter the number of samples in the yellow cell, number of genes in the green one, and prepare the qPCR mixes accordingly as follows:

Table14					
	A	B	C	D	E
1		10uL			
2	Reagent	Volume (ul) x1	Volume (uL) (triplicate of all samples +extra 3 wells)	Number of samples	Number of genes to look at
3	Water, PCR grade (vial 2, colorless cap)	2.5	240	31	4
4	PCR Primer mix 10x (5uM)	1	96	Total ul needed (MM)	Total ul needed (H2O)
5	Master Mix, 2x conc. (vial 1, green cap)	5	480	1920	960
6	Total volume	8.5	816	102	

QPCR materials:

- a. Greiner Sapphire PCR comsumable (384 Well PCR Microplates)

- b. Film for 384 well plate (not aluminum or breathwasy film)
- c. 8-channel (12-channel) auto-multichannel pipette (Boeke lab)
- d. Reagents (RT-PCR box in -20A)
 - I. 2x master mix
 - II. nuclease-free water
- e. Tips for multichannel (small, thin ones)
- f. 25mL resevoir
- g. primers (1:10) from the stock
(primer mixes are in the freezer in TC under bench)

Procedure:

- a. load DNA with pipette setting of 8-channel auto-multichannel pipette from 8-tube strip
 - I. set up tip spacing: 3 steps -> 4.5mm, 9mm, 13.5mm
 - II. pipette volume: 2uL
 - III. save
 - IV. Aspirate with 9mm and dispense with 13.5mm (it's three replicate/sample). Change tips every row.
- b. transfer primer master mix into 4*8-tube strip.
- c. Add 8.5 uL master mix for each gene. Aspirate with 9mm and dispense with 4.5mm by column. (the volume is just enough)
 - I. setting: pipette -> edit -> adjust volume
- d. Run the default on LC480
 - I. make sure the computer of **Felix&LC480** computer is on
 - II. press the corresponding bottom of the **Felix&LC480** computer (e.g., 4)
 - III. log into NYUMC account (Boeke lab account)
 - IV. Open LightCycler480 software
 - V. Login:
 - Username: admin
 - Password: LightCycler480
 - VI. **Put plate in machine by pressing the right botton (If the machine has red light -> no plate inside. The "Start Run" button won't be available until you put plate into the machine)**
 - VII. Run from template like SYBR Green I
 - detection format: SYBR Green I/HRM Dye
 - check block size: 384
 - volume: 10
 - VIII. Save experiment
 - IX. Start run
- e. Export table from LightCycler480
 - I. click Analysis Tab
 - II. create new analysis: Abs Quant/Fit Points or Abs Quant/**2nd Derivative Max**
 - III. select all sample
 - IV. click Calculate botton at the bottom to calculate Cp number
 - V. Export the table as .txt

Analyze the experiment with this [spreadsheet](#)

RT qPCR

Introduction

Freezing of cells, RNA extraction, RT and qPCR protocol with hMNs

Materials

- › hMNs in culture, day 10 after plating
- › ice cold PBS
- › RNA mini Kit Invitrogen
- › Needles 18G
- › 1 mL syringe
- › Maxima™ H Minus cDNA Synthesis Master Mix Catalog number: M1681
- › Integra Biosciences Corp 3423, 125UL 10ECO RKS, 384TIPS
- › Roche Diagnostics LIGHTCYCLER 480 SYBR GREEN I M
- › Sapphire Microplate 384 well for qPCR

›

Procedure

Freezing of cells in plate

- ✓ 1. Once the treatment is done, remove medium from the cells
- ✓ 2. Optional: ice cold PBS wash (hMNs are very fragile, I usually avoid washing to make sure not to lose them)
- ✓ 3. Freeze the plate in liquid nitrogen and transfer to -80 until extraction.

RNA extraction

- ✓ 4. Follow RNA mini Kit Invitrogen user manual
- ✓ 5. Only change: added a DNaseI treatment! Do a first wash with Wash Buffer I with 350 ul. Then incubate 15' at RT with 10ul DNaseI in 80 ul water per sample. Then spin and another 350ul Wash Buffer I.
- ✓ 6. Elute in 50 ul and quantify with Nanodrop.

RT

- ✓ 7. Only maximum 10 ul can be used for the following step, so the RNA concentration is what limits your choice for RNA amount to start with. I have used even 50 ng total to start with success.
- ✓ 8. DNase treatment in PCR tube. If not process with dsDase, use 5uL RNA template and do step 14 directly.

Table1 ^

	A	B	C
1		standard	saving
2	10X dsDNase Buffer	1 µL	0.5
3	dsDNase	1 µL	0.5
4	Template RNA •Total RNA or •poly(A) mRNA or •Specific RNA	1 pg to 5 µg 0.1 pg to 500 ng 0.01 pg to 500 ng	
5	Water, nuclease-free	to 12 µL	to 5uL
6	Total volume	12 µL	5uL

- ✓ 9. Prepare mix of
 - 1ul DNase/sample +1
 - 1ul of buffer 10x/sample +1
- ✓ 10. Add 2 ul/ tube.
- ✓ 11. Mix gently and centrifuge.
- ✓ 12. Incubate at 37°C in a preheated thermomixer or water bath for 2 minutes.
- ✓ 13. Chill on ice, briefly centrifuge and place on ice.
- ✓ 14. Add the following components to the tube on ice:

Table2

	A	B	C
1		standard	saving
2	Maxima cDNA H Minus Synthesis Master Mix (5X)	4 µL	2uL
3	Water, nuclease-free	4 µL	3uL



- ✓ 15. Prepare mix of
 - 4ul Maxima cDNA H Minus Synthesis Master Mix (5X)/sample +1
 - 4ul of Water, nuclease-free/sample +1
- ✓ 16. Add 8 ul per tube.
- ✓ 17. Mix gently and centrifuge. Load the program in the PCR machine:

RT program

	A	B
1	temp	time
2	25	10
3	50	15
4	85	5



- ✓ 18. Incubate at 25°C for 10 minutes.
- ✓ 19. Incubate at 50°C for 15 minutes.
- ✓ 20. Terminate the reaction by heating at 85°C for 5 minutes.
- ✓ 21. Use directly or freeze in -80 for long term storage.

qPCR

- ✓ 22. Dilute the cDNA 1:5 with nuclease-free water, to have 120 ul/sample total. You will use 3 ul of DNA/qPCR reaction.
 - If do "Saving" protocol, dilute cDNA 1:3. Add **2uL** DNA/qPCR reaction.
- ✓ 23. Prepare the table to guide the loading of the qPCR.
- ✓ 24. Use this as template:

wt-11 BP

wt-11 1h Hypo

wt-11 24h Hypo

Table3																								
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
2	ELAV3-	ELAV3-	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	ELAV3	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-	cry_ELAV3	cry_ELAV3-	cry_ELAV3-	cry_ELAV3-						
3	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP	cry_PFKP						
4	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	STM2-	cry_STMN2	cry_STMN2	cry_STMN2	cry_STMN2	cry_STMN2	cry_STMN2	cry_STMN2	cry_STMN2	cry_STMN2						
5	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	GAPDH	PFKP	cry_PFKP	ACTB												
6	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	ACTB	STM2-	cry_STMN2													
7																								
8																								
9																								
10																								
11																								
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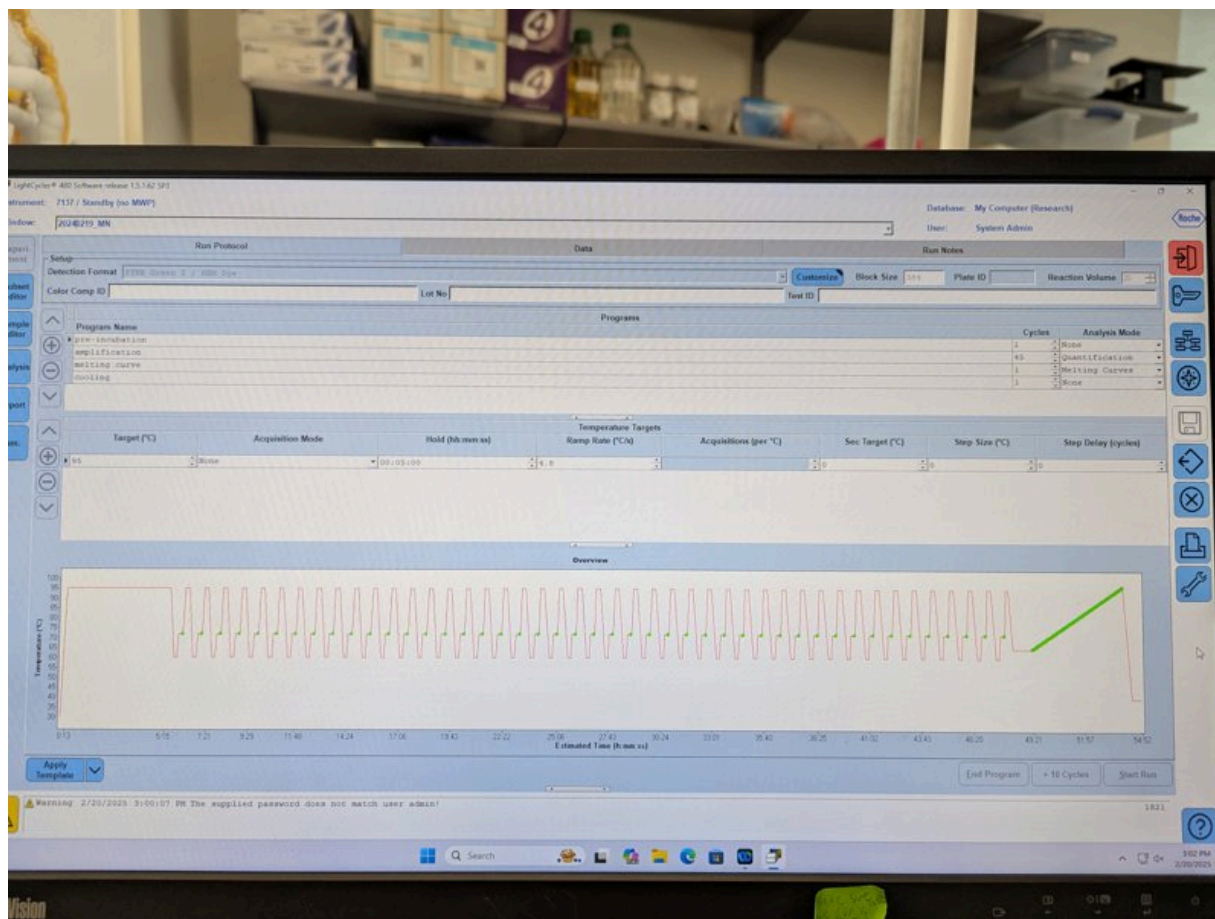
✓ 25. Enter the number of samples in the yellow cell, number of genes in the green one, and prepare the qPCR mixes accordingly as follows:

standard					
	A	B	C	D	E
1	Reagent	Volume (ul) x1	Volume (uL) (triplicate of all samples +extra 3 wells)	Number of samples	Number of genes to look at
2	Water, PCR grade (vial 2, colorless cap)	5	255	16	4
3	PCR Primer mix 10x (5uM)	2	102	Total ul needed (MM)	
4	Master Mix, 2x conc. (vial 1, green cap)	10	510	2040	
5	Total volume	17	867		

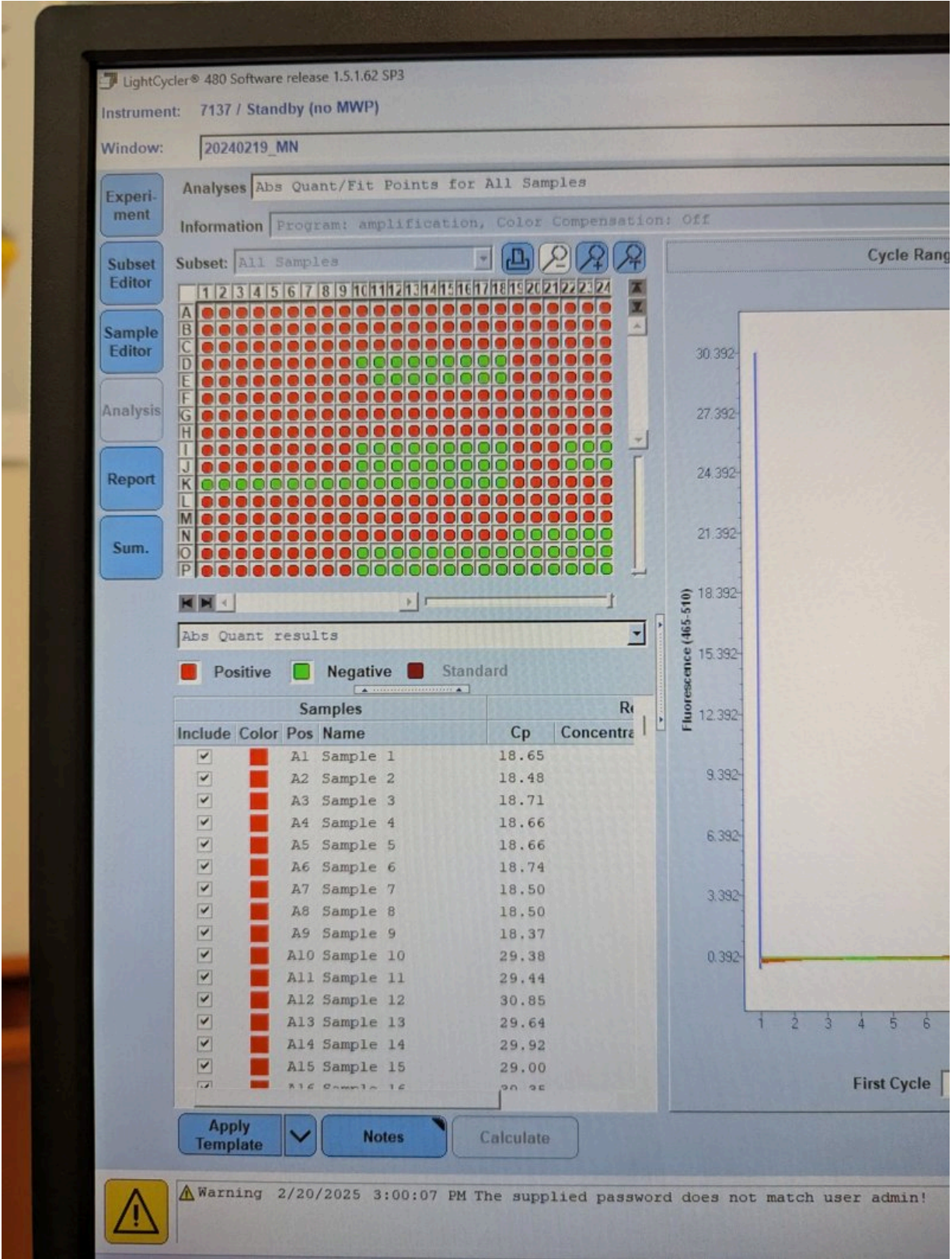
saving										
	A	B	C	D	E	F	G	H	I	J
1		10uL					7uL			
2	Reagent	Volume (ul) x1	Volume (uL) (triplicate of all samples +extra 3 wells)	Number of samples	Number of genes to look at		Volume (ul) x1	Volume (uL) (triplicate of all samples +extra 3 wells)	Number of samples	Number of genes to look at
3	Water, PCR grade (vial 2, colorless cap)	2.5	127.5	16	4		1	51	16	4
4	PCR Primer mix 10x (5uM)	1	51	Total ul needed (MM)			0.5	25.5	Total ul needed (MM)	
5	Master Mix, 2x conc. (vial 1, green cap)	5	255	1020			3.5	178.5	714	
6	Total volume	8.5	433.5				5	255		

- ✓ 26. **Prepare QPCR materials:**
 - Greiner Sapphire PCR consumable (384 Well PCR Microplates)
 - Film for 384 well plate (not aluminum or breathwasy film)
 - 8-channel auto-multichannel pipette (Boeke lab)
 - Reagents (RT-PCR box in -20A)
 - 2x master mix
 - nuclease-free water
 - Tips for multichannel (small, thin ones)
 - 25mL resevoir
 - primers (1:10) from the stock (primer mixes are in the freezer in TC under bench)
- ✓ 27. **prepare primer mix as table above**
- ✓ 28. **load DNA and primer mix as below**
- ✓ 29. setup tip spacing of 8-channel auto-multichannel pipette to 4.7 for 384 wells
- ✓ 30. load DNA with repeat dispense setting of 8-channel auto-multichannel pipette
 - dispense: 3uL
 - first/last: 3uL
 - count: 14
 - save
 - run for every dispense
 - load until last one
 - purge run
- ✓ 31. add 17uL master mix for each gene with 8-channel auto-multichannel pipette
 - setting: pipette -> edit -> adjust volume
- ✓ 32. **Run the default on LC480 as below**
- ✓ 33. make sure the computer of **Felix&LC480** computer is on
- ✓ 34. press the corresponding bottom of the **Felix&LC480** computer (e.g., 4)
- ✓ 35. log into NYUMC account (Boeke lab account)
- ✓ 36. Open LightCycler480 software
- ✓ 37. Login:
 - Username: admin
 - Password: LightCycler480
- ✓ 38. Put plate in machine by pressing the right botton

- ✓ 39. Run from template like SYBR Green I
 - detection format: SYBR Green I/HRM Dye
 - check block size: 384
 - volume: 20 (If use saving, change to the volume you used)



- ✓ 40. Save experiment
- ✓ 41. Start run
- ✓ 42. Export table from LightCycler480 as below
- ✓ 43. click Analysis Tab
- ✓ 44. create new analysis: Abs Quant/Fit Points or Abs Quant/2nd Derivative Max
- ✓ 45. select all sample
- ✓ 46. click **Calculate** button at the bottom to calculate Cp number



✓ 47. Export the table as .txt

✓ 48. Analyze the experiment. You can use [this table](#) as a template.